

Name :- N. Punith Kumar

Reg NO:- 192525156

course code:- CSA0307

course name:- data structure.

CO2-AT1- Analytical problem solving

1. circular queue (size=6)

Initial queue

Index	0	1	2	3	4	5
Value	-	-	-	-	-	-

front = -1, Rear = -1

operation 1:- Enqueue(10)

110|-|-|-|-|-

front=0, Rear=0

operation 2:- Enqueue(20)

110|20|-|-|-|-

front=0, Rear=1

operation 3:- Enqueue(30)

110|20|30|-|-|-

front=0, Rear=2

operation 4:- Dequeue()

10 is removed

1|-|20|30|-|-|-

front=1, Rear=2

operation 5:- Enqueue(40)

1|-|20|30|40|-|-

front=1, Rear=3

operation 6:- Enqueue(50)

1|-|20|30|40|50|-

front = 1, Rear = 4

operation 7: enqueue (60)

1-1201301401501601

front = 1, Rear = 5

operation 8:- enqueue (70)

Rear wraps around to index 0.

1701201301401501601

front = 1, Rear = 0

Queue order:-

20 → 30 → 40 → 50 → 60 → 70

Final Answer

operation	front	Rear
Initial	-1	1
enqueue(10)	0	0
enqueue(20)	0	1
enqueue(30)	0	2
dequeue()	1	2
enqueue(40)	1	3
enqueue(50)	1	4
enqueue(60)	1	5
enqueue(70)	1	0

2) Dynamic selection B/w Linear search and Binary search.

Decision criteria.

The system chooses the search algorithm using:

- Data size (N)
- Sortedness (S)
- Frequency of queries (Q)

Mathematical Analysis:-

case 1: Data is unsorted

Binary search cannot be applied
use Linear search.

Time complexity:-

$$T_L = O(N)$$

case 2:- Data is sorted
compare both algorithms.

Linear search:

$$T_B = O(\log_2 N)$$

choose Binary search if:-

$$\log_2 N < N$$

which is true for all practical values of N .

considering query frequency.

If the dataset is unsorted & there are many searches. First sort it.

Total cost:

$$T = O(N \log N) + Q \times O(\log N)$$

without sorting:

$$T = Q \times O(N)$$

choose Binary search when:

$$N \log N + Q \log N < QN$$

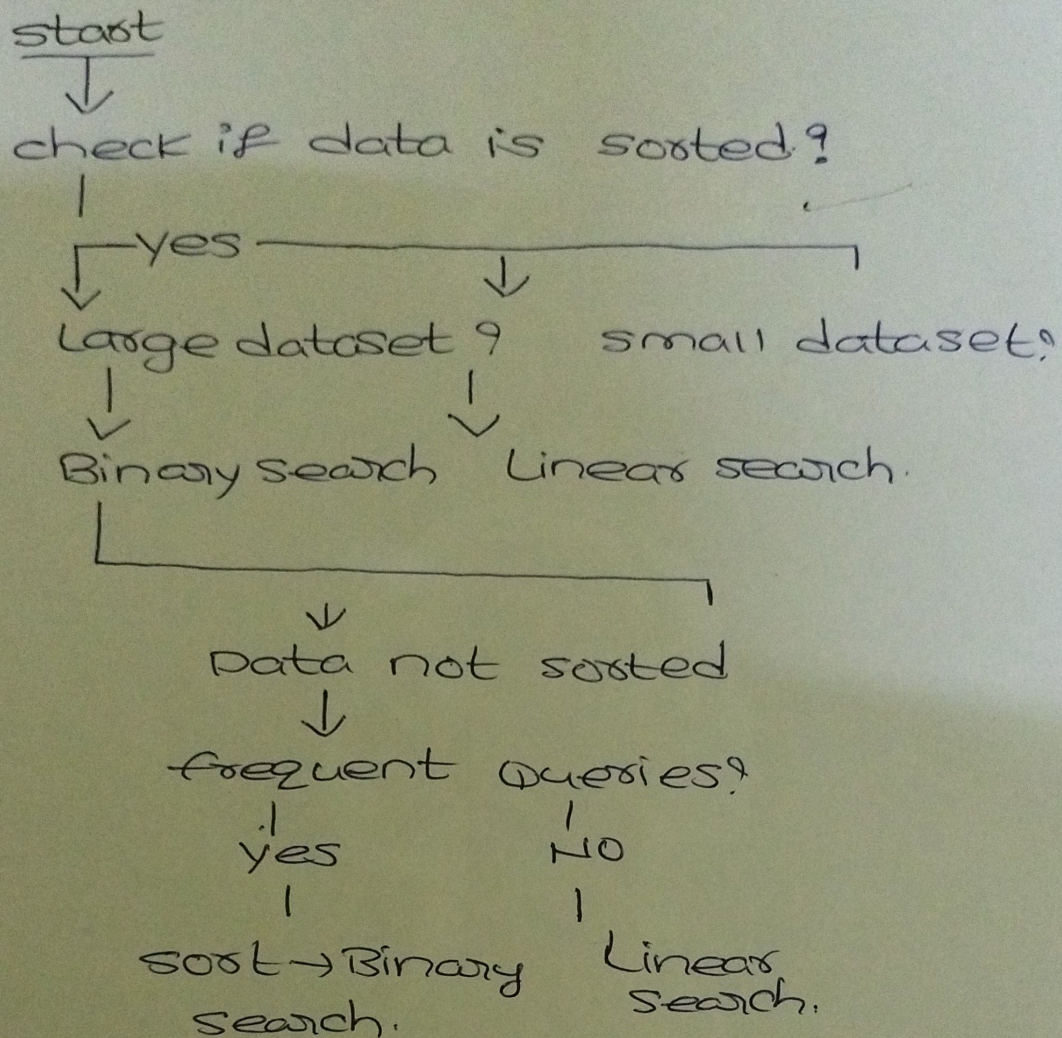
or

$$Q > \frac{N \log N}{N - \log N}$$

Decision Table

condition	Algorithm
unsorted data	Linear search.
sorted data	Binary search.
small dataset ($N < 20$)	Linear search.
Large sorted data - set	Binary search.
frequent queries	sorte Once + Binary search.
one-time search.	Linear search.

flow of the system



Time complexity summary

Algorithm	Best	Average	worst
Linear search	$O(1)$	$O(N)$	$O(N)$
Binary search	$O(1)$	$O(\log N)$	$O(\log N)$